P1 Setup Guide: Python & Jupyter for FIT1061

This guide walks you through setting up your computer for FIT1061. By the end, you'll have Python, Jupyter, and all the libraries you need for the entire semester.

Estimated time: 15–30 minutes (depending on your internet speed).

If you get stuck: Don't spend more than 20 minutes on any single step. Come to your first lab session and a tutor will help you. There is also a cloud fallback (Google Colab) at the end of this guide if nothing else works.

Already have Python or Anaconda installed?

Many of you will have Python on your machine from FIT1045 or another course. That's fine: read the scenario that matches you.

How to check what you have

Open a terminal (Windows: Command Prompt or PowerShell; macOS/Linux: Terminal) and run these commands. It's OK if some of them fail: that just tells you what you *don't* have.

```
python --version
conda --version
```

(On macOS/Linux, try `python3 --version` if `python` doesn't work.)

Scenario A: "I already have Anaconda or Miniconda"

You'll see something like `conda 26.x.x` when you run `conda --version`.

You don't need to install anything. Skip straight to **Step 3** (download task resources) and **Step 4** (create the FIT1061 environment). Your existing conda installation will manage the new environment alongside any others you have. They don't interfere with each other.

If your conda is very old (below version 23), update it first:

```
conda update conda
```

Scenario B: "I have Python from python.org, Homebrew, or FIT1045, but no conda"

You'll see a Python version (e.g., `Python 3.11.5`) but `conda --version` will fail.

Don't uninstall your existing Python. Install Miniconda alongside it. The conda environment system is completely separate: when you activate the `fit1061` environment, it uses its own Python, ignoring whatever else is on your system.

Start from **Step 1** and follow the full guide.

Scenario C: "I have no idea what's installed"

If both commands fail or you're not sure, start from **Step 1**. The guide assumes nothing.

The key thing to understand

Conda environments are **isolated**. When you type `conda activate fit1061`, your terminal switches to a self-contained Python setup that has exactly the libraries FIT1061 needs. It doesn't touch, change, or break anything else on your computer. When you type `conda deactivate`, you're back to whatever you had before.

This means: - Other courses' Python setups won't break FIT1061. - FIT1061 won't break other courses' setups. - If something goes wrong, you can delete the environment and recreate it without affecting anything else: `conda env remove -n fit1061  conda env create -f fit1061_environment.yml`

What are we installing?

Before we start, here's what each piece does:

- **Miniconda**: A small program that manages Python and its libraries for you. Think of it as an app store specifically for Python tools. It handles installing, updating, and keeping everything compatible so you don't have to.
- **Python**: The programming language we use all semester.
- **Jupyter Notebook**: A special editor that lets you write Python code in "cells" and run them one at a time, seeing results immediately. It runs in your web browser but works on your own computer.
- **NumPy**: A library for working with arrays of numbers efficiently. We use it from Week 6 onwards.
- **Matplotlib**: A library for making graphs and plots. We use it to visualise algorithm behaviour.

You don't need to install these one by one. Miniconda will install everything for you in one step using an **environment file** we provide.

Step 1: Download Miniconda

Go to the Miniconda download page: **https://docs.anaconda.com/miniconda/**

Download the installer for your operating system. **Choose the one that matches your computer:**

Windows

1. Click the **Windows** installer (the `.exe` file).
2. It will usually be called something like `Miniconda3-latest-Windows-x86_64.exe`.
3. Save it anywhere (your Downloads folder is fine).

macOS

1. If you have an **Apple Silicon Mac** (M1, M2, M3, M4, or any Mac bought after late 2020), click the **macOS Apple M1 64-bit** installer (the `.pkg` file).
2. If you have an **older Intel Mac**, click the **macOS Intel x86 64-bit** installer (the `.pkg` file).
3. **Not sure which one you have?** Click the Apple menu () in the top-left corner of your screen → "About This Mac". If it says "Apple M1" or "Apple M2" (or similar), you have Apple Silicon. If it says "Intel", you have Intel.

Linux

1. Click the **Linux 64-bit** installer (the `.sh` file).
2. Save it to your Downloads folder.

Step 2: Install Miniconda

Windows

1. **Double-click** the downloaded `.exe` file to start the installer.
2. Click **Next** on the welcome screen.
3. Click **I Agree** on the licence screen.
4. Choose **Just Me** (recommended) and click **Next**.
5. Leave the install location as the default and click **Next**.
6. On the "Advanced Options" screen:

- **Check** "Add Miniconda3 to my PATH environment variable". (The installer says it's not recommended; ignore that. For this course, having it on PATH makes everything easier.)
- **Check** "Register Miniconda3 as my default Python".
- Click **Install**.

7. Wait for the installation to finish. Click **Finish**.

To check it worked:

1. Open the **Start Menu** and search for **"Anaconda Prompt"** (not the regular Command Prompt).
2. Click to open it. You should see a black window with text like `(base) C:\Users\YourName>`.
3. Type `conda --version` and press Enter.
4. You should see something like `conda 24.x.x`. If you do, Miniconda is installed.

What is the Anaconda Prompt? It's a special command-line window that knows where Miniconda is installed. On Windows, always use the Anaconda Prompt (not the regular Command Prompt or PowerShell) when running conda commands. This avoids PATH issues.

macOS

1. **Double-click** the downloaded `.pkg` file.
2. Follow the installer prompts: click **Continue**, **Agree**, **Install**.
3. You may need to enter your password (the one you use to log into your Mac).
4. Wait for the installation to finish. Click **Close**.

To check it worked:

1. Open **Terminal**. (You can find it by pressing `Cmd + Space`, typing "Terminal", and pressing Enter.)
2. **Close and reopen Terminal** if it was already open: this is important so it picks up the new installation.
3. Type `conda --version` and press Enter.
4. You should see something like `conda 24.x.x`. If you do, Miniconda is installed.

If you see "conda: command not found": Close Terminal completely and reopen it. If it still doesn't work, try this: type `source ~/miniconda3/bin/activate` and press Enter, then try `conda --version` again. If that works, run `conda init zsh` (or `conda init bash` if you use bash) and restart Terminal.

Linux

1. Open a **terminal**.
2. Navigate to your Downloads folder: `cd ~/Downloads`
3. Make the installer executable: `chmod +x Miniconda3-latest-Linux-x86_64.sh`
4. Run it: `./Miniconda3-latest-Linux-x86_64.sh`
5. Press Enter to read the licence, then type `yes` to accept.
6. Press Enter to accept the default install location.
7. When asked "Do you wish to update your shell profile to automatically initialize conda?" type **yes**.
8. **Close and reopen your terminal.**

To check it worked:

1. Type `conda --version` and press Enter.
2. You should see something like `conda 24.x.x`.

Step 3: Download the task resources

Download the P1 task resources ZIP from OnTrack and unzip it into a folder you can find easily. For example:

- **Windows:** `C:\Users\YourName\Documents\FIT1061\P1\`
- **macOS / Linux:** `~/Documents/FIT1061/P1/`

After unzipping, you should see these files:

```
P1/
  P1_hello_ai.ipynb
  P1_learning_plan_template.ipynb
  P1_python_bridge.ipynb
  fit1061_environment.yml
  P1_setup_guide.md          (this file)
```

Step 4: Create the FIT1061 environment

This is the step where we install Python, Jupyter, NumPy, and Matplotlib all at once.

What is an "environment"?

An environment is a self-contained Python setup. It has its own version of Python and its own libraries, separate from anything else on your computer. This means:

- FIT1061 libraries won't interfere with other courses or projects.
- If something goes wrong, you can delete the environment and start fresh without affecting anything else.
- Everyone in FIT1061 has the same setup, so code that works for your classmate will work for you.

Creating the environment

1. Open your terminal:
   - **Windows:** Open **Anaconda Prompt** (search for it in the Start Menu).
   - **macOS / Linux:** Open **Terminal**.
2. Navigate to the folder where you unzipped the task resources. For example:
   - **Windows:** `cd C:\Users\YourName\Documents\FIT1061\P1`
   - **macOS / Linux:** `cd ~/Documents/FIT1061/P1`
3. Run this command:

   ```
   conda env create -f fit1061_environment.yml
   ```

4. **Wait.** This downloads and installs everything. It may take 2–10 minutes depending on your internet speed. You'll see lots of text scrolling by; that's normal.
5. When it finishes, you should see a message like:

   ```
   done
   #
   # To activate this environment, use
   #
   #     $ conda activate fit1061
   ```

   If you see an error about "solving environment": This usually means a slow internet connection or a temporary server issue. Wait a minute and try again. If it keeps failing, try: `conda env create -f fit1061_environment.yml --channel conda-forge`

Step 5: Activate the environment and open Jupyter

Every time you work on FIT1061, you need to **activate** the environment first. This tells your computer "use the FIT1061 Python setup, not the default one."

1. In your terminal (Anaconda Prompt on Windows), type:

   ```
   conda activate fit1061
   ```

2. Your prompt should change. You'll see `(fit1061)` at the beginning of the line instead of `(base)`:

   ```
   (fit1061) C:\Users\YourName\Documents\FIT1061\P1>
   ```

3. Now start Jupyter Notebook:

   ```
   jupyter notebook
   ```

4. **Your web browser will open automatically** with the Jupyter file browser. You should see your P1 files listed.
5. Click on **`P1_hello_ai.ipynb`** to open the Hello AI notebook.
6. Run every cell in the notebook (click each cell and press `Shift+Enter`).

   "jupyter notebook" opens a browser tab: is that right? Yes! Jupyter runs as a local web server on your computer. The browser is just the interface. Your files stay on your computer, not in the cloud.

   How do I stop Jupyter when I'm done? Go back to your terminal and press `Ctrl+C` twice. Or just close the terminal window.

Step 6: Verify your notebook before submitting

After completing the Hello AI notebook:

1. In Jupyter, choose **Kernel → Restart and Run All**.
2. Confirm every cell finishes without error and the output you expect appears under each task.
3. **Save the notebook** (Ctrl+S / Cmd+S).
4. Upload to OnTrack and mark as **Ready for Feedback**.

If a cell errors, fix it, restart + run all again, then save.

Troubleshooting

"conda: command not found"

- **Windows:** Make sure you're using the **Anaconda Prompt**, not the regular Command Prompt or PowerShell.
- **macOS / Linux:** Close your terminal completely and reopen it. If still not found, run `source ~/miniconda3/bin/activate` and then `conda init zsh` (or `conda init bash`). Restart terminal.

"conda activate fit1061" gives an error

- Make sure you ran `conda env create -f fit1061_environment.yml` first (Step 4).
- Check for typos: the environment is called `fit1061` (no spaces, no capitals).
- Try: `conda env list`. You should see `fit1061` in the list. If not, Step 4 didn't complete successfully. Try it again.

Jupyter doesn't open in the browser

- Try opening your browser manually and going to: `http://localhost:8888`
- If that doesn't work, look at the terminal output: Jupyter prints a URL with a token. Copy and paste the full URL into your browser.

"ModuleNotFoundError: No module named 'numpy' "

- You probably forgot to activate the environment. Run `conda activate fit1061` first, then start Jupyter again.

The plot in Step 4 of the notebook doesn't show

- Make sure you're running Jupyter Notebook (not JupyterLab, which sometimes needs a different renderer). The command `jupyter notebook` should work.
- Try adding `%matplotlib inline` at the top of the cell and running it again.

My computer is very old / I have a Chromebook

- Skip to the **Google Colab fallback** section below.

Nothing is working and I've been stuck for 20 minutes

- **Stop.** Come to your first lab session. Your tutor will help you set up.
- In the meantime, use **Google Colab** (below) so you can start working on the other P1 sub-tasks.

Fallback: Google Colab

If you cannot get Miniconda working on your machine, you can use Google Colab as a temporary solution. Colab runs Jupyter notebooks in the cloud, no installation needed.

1. Go to **https://colab.research.google.com** (you need a Google account).
2. Click **File → Upload notebook**.
3. Upload `P1_hello_ai.ipynb` from your P1 folder.
4. Run every cell as normal.

Important: - Colab already has Python, NumPy, and Matplotlib installed, so the setup check cells will work. - Colab is a **temporary fallback**, not a replacement. You should get Miniconda working on your own machine for the rest of the semester. Some later tasks involve local files that are easier to manage locally. - To download your completed notebook from Colab: **File → Download → Download .ipynb**.

Quick reference: the commands you'll use every week

Every time you sit down to work on FIT1061:

```
conda activate fit1061
cd path/to/your/FIT1061/folder
jupyter notebook
```

That's it. Three commands and you're working.

FIT1061 Setup Guide: Introduction to Artificial Intelligence